

- 3 People living in some villages in Ecuador, a South American country, keep chickens as a source of food.

These people have been encouraged to add ampicillin, an antibiotic, to the chicken food.

Feeding ampicillin to chickens may cause the bacterium, *E. coli*, to become resistant to ampicillin.

Village **A** was the first in the area to add ampicillin to the chicken food.

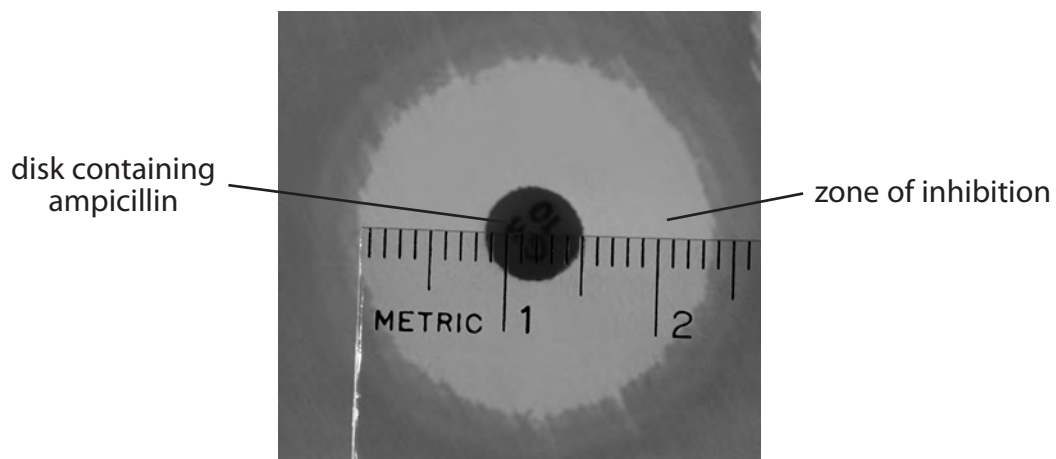
Village **B** has only recently added ampicillin to the chicken food.

Samples of *E. coli* were taken from the guts of chickens from each village.

Each sample was cultured on a nutrient agar plate.

A disc containing ampicillin was placed on each culture. After incubation, the diameter of the clear zone of inhibition was measured in millimetres.

The photograph shows the zone of inhibition on one nutrient agar plate after incubation for 72 hours.



(Source: © Gado Images / Alamy Stock Photo)

The results:

Village A

15.9 16.2 14.5 17.1 15.3 15.4 16.2 16.8 14.4 16.8
18.2 16.1 16.9 16.6 16.4

Village B

17.2 17.0 17.3 15.8 17.3 18.9 19.1 19.3 16.9 14.8
16.7 17.4 17.8 17.4 16.8

(a) State a suitable null hypothesis for this investigation.

(1)

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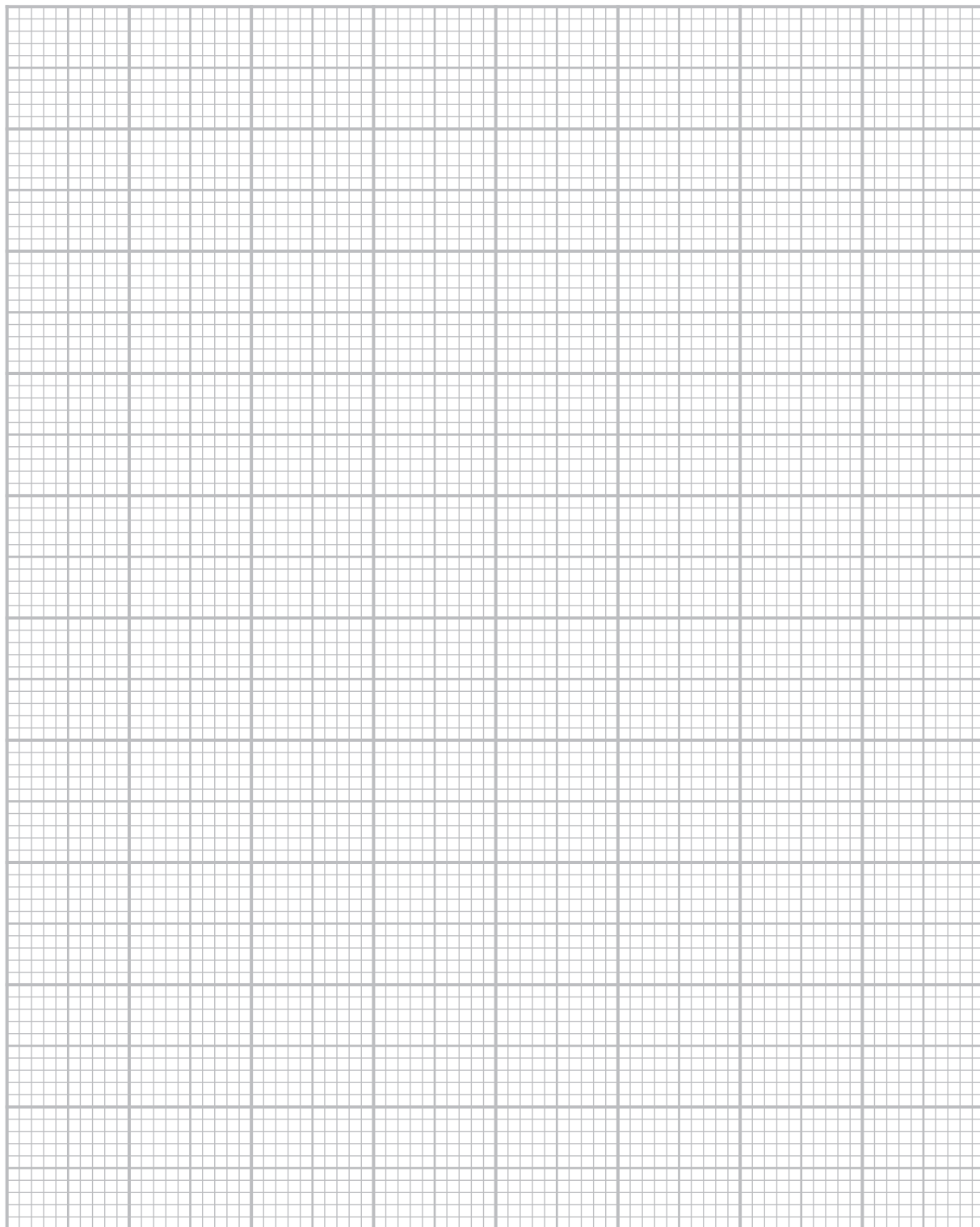
(b) (i) Draw a suitable table to display the results and your calculated **means** for village **A** and for village **B**.

(3)

- (ii) Plot a graph to show the mean diameter of the zone of inhibition for village **A** and for village **B**.

Include an indication of the variability of the results.

(3)



(iii) A student analysed the data with a t test using the formula

$$t = \frac{(\bar{x}_A - \bar{x}_B)}{\sqrt{\frac{(S_A)^2}{n_A} + \frac{(S_B)^2}{n_B}}}$$

where:

$\bar{x}$ is the mean value for each village

n is the number of samples for each village

$(S_A)^2 = 0.99$ and $(S_B)^2 = 1.39$

Calculate the value of t .

(3)

Answer

(iv) The table shows the critical values of t for different degrees of freedom.

The number of degrees of freedom = $(n_A - 1) + (n_B - 1)$

Degrees of freedom	p = 0.05	p = 0.01
15	2.13	2.95
16	2.12	2.92
17	2.11	2.90
18	2.10	2.88
19	2.09	2.86
20	2.09	2.84
21	2.08	2.83
22	2.07	2.82
23	2.07	2.81
24	2.06	2.80
25	2.06	2.79
26	2.06	2.78
27	2.05	2.77
28	2.05	2.76
29	2.04	2.76
30	2.04	2.75

A scientist suggested that there was evidence for the development of resistance to ampicillin in *E. coli* in Village **B**.

Criticise this suggestion.

Use the information given and the analysis of the results to support your answer.

(2)

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(c) Suggest **two** improvements that could be made to this investigation.

(2)

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(Total for Question 3 = 14 marks)